

HORNSBY GIRLS HIGH SCHOOL



2009 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time- 5 minutes
- Working Time – 2 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown for every question
- Begin each question on a fresh sheet of paper

Total marks (84)

- Attempt Questions 1-7
- All questions are of equal value

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Total Marks**Attempt Questions 1–7****All Questions are of equal value**

Begin each question on a NEW SHEET of paper, writing your student number and question number at the top of the page. Extra paper is available.

Question 1 (12 marks) Use a SEPARATE sheet of paper.	Marks
(a) Using the table of standard integrals, evaluate $\int \frac{1}{\sqrt{x^2 + 16}} dx$.	1
(b) If $f(x) = \cot x$, find $f'(x)$ in simplest form.	2
(c) Find the acute angle between the lines $y = 2x + 1$ and $3x + y - 7 = 0$.	2
(d) Write $P(x) = x^3 - 6x^2 + 3x + 1$ in the form $P(x) = Q(x) \times T(x) + c$ where $T(x) = x - 1$ and c is a constant.	2
(e) Find the vertical and horizontal asymptotes of $y = \frac{3x^3}{x^3 + 8}$.	2
(f) Evaluate $\int_0^1 \sqrt{1 - x^2} dx$ using the substitution $x = \sin \theta$.	3

Question 2 (12 marks) Use a SEPARATE sheet of paper.

Marks

- (a) The mathematics staff currently consists of 3 male and 6 female teachers. How many ways can a team of 5 be chosen from the mathematics staff which contains at least 3 female teachers? **2**
- (b) The area bounded by the curve $y = e^x$, the x -axis and the ordinates at $x = 0$ and $x = 3$ is rotated about the x -axis. Use Simpson's rule with 3 function values to find an approximation of the volume so generated. **3**
- (c) Determine the coordinates of the point Q if it divides the interval joining $P(-3, 1)$ and $R(5, -10)$ externally in the ratio $3:5$. **2**
- (d) By sketching $y = \cos x + \sin x$ and $y = x$ for $0 \leq x \leq \frac{\pi}{2}$, find the number of solutions of $\cos x + \sin x - x = 0$ in the domain $0 \leq x \leq \frac{\pi}{2}$. **2**
- (e) The probability that a vaccine for a new virus will not protect the patient from getting the virus is found to be 0.02%. If a group of 1000 people are vaccinated, find the probability, as a percentage, that:
- (i) all the group will be protected from getting the virus. **1**
- (ii) at most 2 people will not be protected from getting the virus. **2**

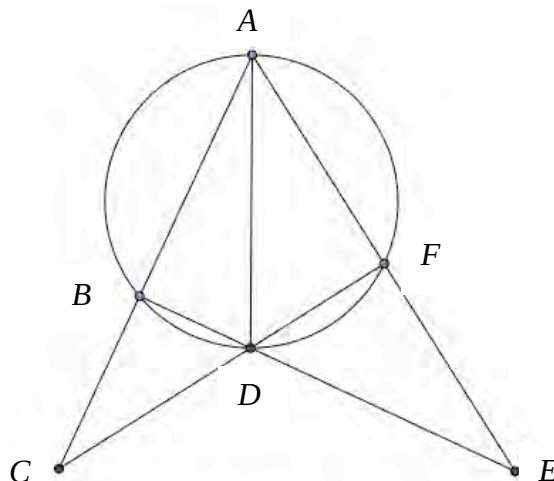
Question 3 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) Solve for x : $\frac{5}{x-1} < 3$

2

(b)



In the diagram above AE , BE , CF & CA are secants and $\angle ACD = \angle AED$

(i) Show that $\angle CBD = \angle EFD$

1

(ii) Prove that AD is a diameter of the circle $AFDB$.

2

(c) Let T be the temperature in a room at time t and let A be the temperature of the room's surroundings. Newton's Law of Cooling states that the rate of change of temperature T is proportional to $(T - A)$.

(i) Verify that $T = A + Be^{kt}$, where B and k are constants, satisfies Newton's Law of Cooling.

1

(ii) The temperature of a substance in a room of constant temperature 6°C is noted to be 29°C .
After 40 minutes the temperature of the substance is noted to be 14°C .
Find how long it takes the temperature of the substance to reach 9°C .
(Give your answer to the nearest minute.)

3

(d) Find the term independent of x in the expansion of $\left(2x^3 - \frac{1}{x}\right)^{12}$.

3

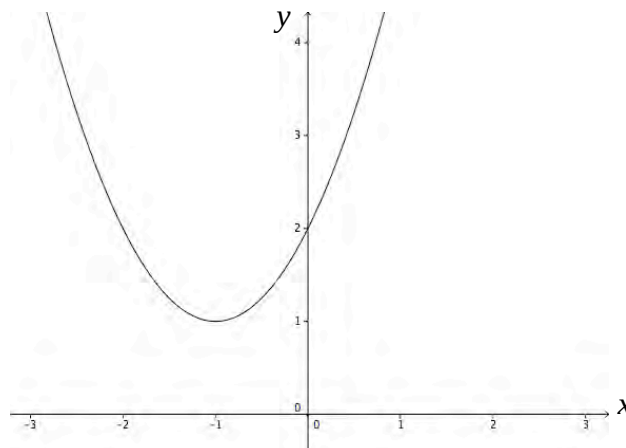
Question 4 (12 marks) Use a SEPARATE sheet of paper.

Marks

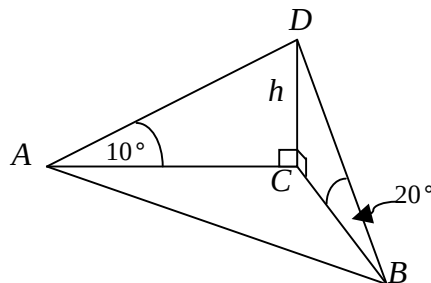
- (a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$.
If $p + q = 8$ find the locus of M , the mid-point of PQ . 2
- (b) Find all the values of θ in the domain $0 \leq \theta \leq 2\pi$ for which
 $3 \cos \theta - 2 \sin \theta = 2$. 3
- (c) (i) Differentiate $x \tan^{-1} x$ 2
- (ii) Hence evaluate $\int_0^1 \tan^{-1} x \, dx$ 2
- (d) If $f(x) = 3x + 5$ and $g(x) = \frac{x-5}{3}$, find $f(g(x) + g(f(x)))$ 3

Question 5 (12 marks) Use a SEPARATE sheet of paper.**Marks**

- (a) The graph of $f(x) = x^2 + 2x + 2$ is shown on the diagram below.



- (i) The function $g(x) = x^2 + 2x + 2$ is defined in the domain $x \geq -1$. 2
Sketch the graph of $g(x)$ and its inverse function $g^{-1}(x)$, showing clearly all intercepts with the coordinate axes.
- (ii) State the domain of $g^{-1}(x)$. 1
- (iii) Find an expression for $y = g^{-1}(x)$ in terms of x . 2
- (b) Use mathematical induction to prove that $4^n \geq 1 + 3n$, for integer $n \geq 1$. 3
- (c) A vertical flagpole CD of height h metres stands with its base C on horizontal ground. A and B are points on the ground where A is due West of C and B is on a bearing of 170° from C . From A and B the angles of elevation of the top D of the flagpole are 10° and 20° respectively.



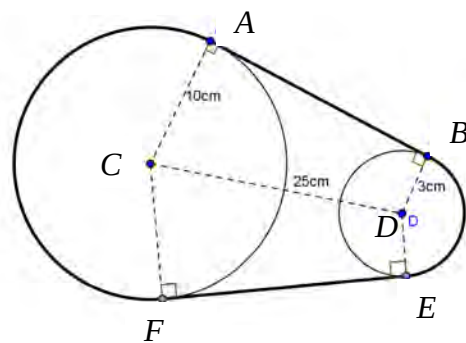
- (i) Express the length of AC in terms of h . 1
- (ii) Show that $\angle ACB$ is 100° . 1
- (iii) If the distance between A and B is 40 metres, find the height of the tower CD . 2

Question 6 (12 marks) Use a SEPARATE sheet of paper.

Marks

- (a) The acceleration $\ddot{x}$ m/s² at time t seconds of a particle moving in a straight line is given by $\ddot{x} = -4 \cos 2t - 8 \sin 2t$.
If the particle is at a distance of x metres from the origin at time t and if initially it is at $x=1$ with a velocity of 4 m/s,
- (i) Show that $\ddot{x} = -4x$. 3
- (ii) Show that the position of the particle after $\frac{\pi}{4}$ seconds is 2 metres to the right of the origin and the magnitude of its velocity is 2m/s at this time. 2
- (iii) Is the speed of the particle increasing or decreasing when $t = \frac{\pi}{4}$? 1
Justify your answer.
- (iv) How long is it exactly between each time the particle passes the origin? 1
Explain your answer.

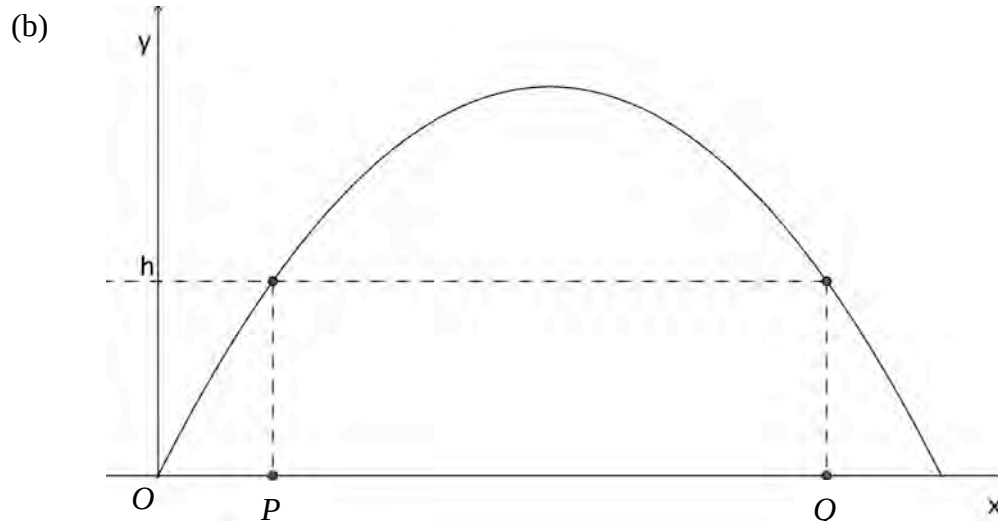
- (b) A belt connects a driving pulley of radius 3cm to another pulley of radius 10cm, as shown in the diagram below:



- (i) Show that $\angle ACD = 73^\circ 44'$ (Hint: Draw a line parallel to AB through D). 1
- (ii) Find the length of AB . 1
- (iii) Find the total length of the belt. 3

Question 7 (12 marks) Use a SEPARATE sheet of paper.**Marks**

- (a) Find the value of n if the coefficient of x^6 is twice the coefficient of x^5 in the expansion of $(3+2x)^n$. 3



A particle is projected with velocity $V \text{ ms}^{-1}$ from a point O at an angle of elevation α . The particle just clears two vertical trees P and Q , of height h metres at horizontal distance of p metres and q metres from O respectively.

The acceleration due to gravity is taken as 10 ms^{-2} and air resistance is ignored.

- (i) Find the Cartesian equation of the flight of the particle. 1
- (ii) Show that $V^2 = \frac{5p^2(1 + \tan^2 \alpha)}{p \tan \alpha - h}$. 2
- (iii) Show that $\tan \alpha = \frac{h(p+q)}{pq}$. 3
- (iv) Another particle was projected with the same angle as the first particle but with velocity $W \text{ ms}^{-1}$ so as to just clear the tree P and land half-way between the trees P and Q . 3

This particle has equations for horizontal and vertical displacement to be $x = Wt \cos \alpha$ and $y = -5t^2 + Wt \sin \alpha$.

Show that $W^2 = \frac{5h(p+q)^2}{pq} + \frac{5pq}{h}$.

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Solution to Q1 Ext 1 2009 Trial

$$(a) \int \frac{1}{\sqrt{x^2+16}} dx = \ln|x + \sqrt{x^2+16}| + C$$

$$(b) f(x) = \cot x$$

$$= \frac{\cos x}{\sin x}$$

$$f'(x) = \frac{\sin x (-\cos x) - \cos x \sin x}{\sin^2 x}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

$$= \frac{-1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x$$

$$(c) y = 2x + 1 \quad y = -3x + 7$$

$$m_1 = 2 \quad m_2 = -3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2 - (-3)}{1 - 6} \right|$$

$$\tan \theta = 1$$

$$\theta = 45^\circ$$

$$(d) P(x) = x^3 - 6x^2 + 3x + 1$$

$$\begin{array}{r} x^2 - 5x - 2 \\ x-1 \overline{) x^3 - 6x^2 + 3x + 1} \end{array}$$

$$\begin{array}{r} x^3 - x^2 \\ \underline{-5x^2 + 3x} \\ -5x^2 + 5x \\ \underline{-2x + 1} \\ -2x + 1 \\ \underline{-2x + 1} \\ -1 \end{array}$$

$$\therefore P(x) = (x-1)(x^2 - 5x - 2) - 1$$

$$(e) y = \frac{3x^3}{x^3 + 8}$$

Horizontal asymptotes:

$$\lim_{x \rightarrow \infty} \frac{3x^3}{x^3 + 8} = \lim_{x \rightarrow \infty} \frac{\frac{3x^3}{x^3}}{\frac{x^3 + 8}{x^3}}$$

$$= \lim_{x \rightarrow \infty} \frac{3}{1 + \frac{8}{x^3}}$$

$$= 3$$

$\therefore y = 3$ is horizontal asymptote.

Vertical asymptote:

$$\text{let } x^3 + 8 = 0$$

$$x = -2$$

$\therefore$ Vertical asymptote is $x = -2$

(f)

$$I = \int_0^1 \sqrt{1-x^2} dx$$

$$\text{let } x = \sin \theta$$

$$\frac{dx}{d\theta} = \cos \theta$$

$$dx = \cos \theta d\theta$$

$$\text{When } x=0, \theta=0$$

$$x=1, \theta = \frac{\pi}{2}$$

$$\therefore I = \int_0^{\pi/2} \sqrt{1-\sin^2 \theta} \cos \theta d\theta$$

$$= \int_0^{\pi/2} \cos^2 \theta d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{2} + \frac{1}{2} \sin \pi \right) - \left(0 + \frac{1}{2} \sin 0 \right) \right]$$

$$= \frac{\pi}{4}$$

Question 2

$$a) 2 \text{ Males at most } \therefore 5F \text{ or } 4F \text{ or } 3F$$

$$\text{OM} \quad \text{IM} \quad \text{2M}$$

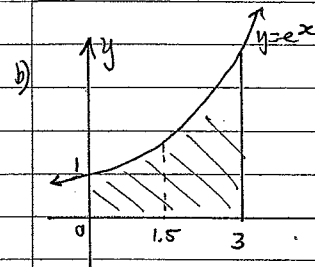
$$\text{Nb. group} = {}^6C_5 x^3 C_0 + {}^6C_4 x^3 C_1 + {}^6C_3 x^3 C_2$$

$$= 6 + 45 + 60$$

$$= 111$$

$$\text{or } {}^9C_5 - {}^3C_3 x^6 C_2 = 126 - 15$$

$$= 111$$



$$V = \pi \int_0^3 (e^x)^2 dx$$

$$= \pi \frac{1}{2} \{ e^0 + 4e^3 + e^6 \}$$

$$= 761.47641 \dots$$

x	0	1.5	3
e^{2x}	1	e^3	e^6

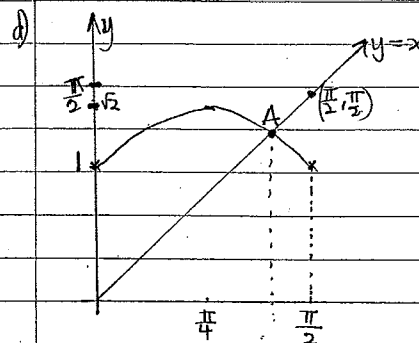
Volume is approximately 761.5 units

$$c) P(-3, 1) \quad R(5, -10)$$

$$\begin{array}{c} \nearrow \\ -3:5 \quad (\text{or } 3:-5) \end{array}$$

$$Q \text{ is } \left(\frac{5 \times -3 + -3 \times 5}{5-3}, \frac{5 \times 1 + -3 \times -10}{5-3} \right)$$

$$= (-15, 17\frac{1}{2})$$



A is where $y = \cos x + \sin x$ and $y = x$ intersect. (1 solution for $0 \leq x \leq \frac{\pi}{2}$)

i.e. where $\cos x + \sin x = x$

$$\cos x + \sin x - x = 0$$

QUESTION 4

$$a) X = \frac{2ap + 2aq}{2}$$

$$= a(p+q)$$

$$\therefore X = 8a$$

$$b) 3\cos\theta - 2\sin\theta = 2$$

$$\sqrt{3^2 + 2^2} = \sqrt{13}$$

$$\tan\alpha = \frac{2}{3}$$

$$\therefore \sqrt{13}\cos(\theta + 0.588) = 2$$

$$\theta + 0.588 = \cos^{-1}\left(\frac{2}{\sqrt{13}}\right)$$

$$\theta = 0.395, 4.712$$

$$\text{OR } \theta = 22.6^\circ, 270^\circ$$

$$\text{OR } \frac{3(1-t^2)}{1+t^2} - \frac{2 \cdot 2t}{1+t^2} = 2$$

$$3 - 3t^2 - 4t = 2 + 2t^2$$

$$5t^2 + 4t - 1 = 0$$

$$(5t-1)(t+1) = 0$$

$$\therefore \tan\theta_1 = \frac{1}{5}, \tan\theta_2 = -1$$

$$\theta = 0.395, \frac{3\pi}{2}$$

$$c) i) \frac{dy}{dx} = \tan^{-1}x + \frac{x}{1+x^2}$$

$$ii) \int \tan^{-1}x \, dx = \left[x \tan^{-1}x \right]_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx$$

$$= \frac{\pi}{4} - \left[\frac{1}{2} \ln(1+x^2) \right]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$$= 0.4388 \dots$$

$$d) g(f(x)) = \frac{3x+5-5}{3}$$

$$= x$$

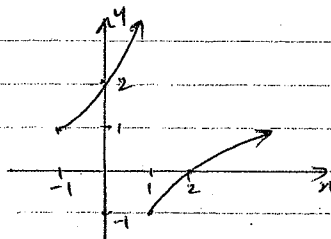
$$g(x) + g(f(x)) = \frac{x-5}{3} + \frac{3x}{3} \quad \therefore f\left(\frac{4x-5}{3}\right) = 3\left(\frac{4x-5}{3}\right) + 5$$

$$= \frac{4x-5}{3}$$

$$= 4x$$

QUESTION 5

a) i)



$$ii) x \geq 1$$

$$iii) x = y^2 + 2y + 2$$

$$x = (y+1)^2 + 1$$

$$y+1 = \sqrt{x-1}$$

$$y = \sqrt{x-1} - 1$$

b) Prove true for $n=1$

$$4 \geq 1+3 \quad \text{True}$$

Assume true for $n=k$

$$4^k \geq 1+3k$$

Prove true for $n=k+1$

$$\text{R.T.D. } 4^{k+1} \geq 3k+4$$

$$\text{now } 4^k \geq 1+3k$$

$$4^{k+1} \geq 4+12k$$

$$\geq 3k+4+9k$$

$$\geq 3k+4$$

$\therefore$ TRUE

etc...

$$c) i) AC = \frac{h}{\tan 10^\circ}$$

$$ii) 270^\circ - 170^\circ = 100^\circ$$

$$iii) AC = \frac{h}{\tan 10^\circ}$$

$$\therefore 1600 = \frac{h^2}{\tan^2 10^\circ} + \frac{h^2}{\tan^2 20^\circ} - \frac{2h^2}{\tan 10^\circ \tan 20^\circ} \cos 100^\circ$$

$$h^2 = \frac{1600 \tan^2 10^\circ \tan^2 20^\circ}{\tan^2 30^\circ + \tan^2 10^\circ - 2 \cos 100^\circ \tan 10^\circ \tan 20^\circ}$$

$$\div 35.458$$

$$\div 35.458$$

$$h = 5.955$$

Q6a) i) $\ddot{x} = -4 \cos 2t - 8 \sin 2t$
 $\dot{x} = \int -4 \cos 2t - 8 \sin 2t \, dt$
 $= -2 \sin 2t + 4 \cos 2t + C$

$t=0, \dot{x}=4: 4 = -2(0) + 4(1) + C$

$\therefore C=0$

$\dot{x} = -2 \sin 2t + 4 \cos 2t$ — ①

$x = \int -2 \sin 2t + 4 \cos 2t \, dt$
 $= \cos 2t + 2 \sin 2t + C_1$

$t=0, x=1: 1 = 1 + 2(0) + C_1$

$\therefore C_1=0$

$x = \cos 2t + 2 \sin 2t$ — ②

$\ddot{x} = -4 \cos 2t - 8 \sin 2t$
 $= -4(\cos 2t + 2 \sin 2t)$
 $= -4x$

ii) $t = \frac{\pi}{4}, x = 2 \cos 2(\frac{\pi}{4}) + 2 \sin 2(\frac{\pi}{4})$

$= 2 \therefore 2 \text{ m to the right of the origin}$

$\dot{x} = -2 \sin 2(\frac{\pi}{4}) + 4 \cos 2(\frac{\pi}{4})$
 $= -2 \text{ ie speed is } 2 \text{ m/s}$

iii) when $t = \frac{\pi}{4}, \ddot{x} = -4(2) = -8 \text{ m/s}^2$

$\therefore$ particle is increasing as accel & velocity have the same sign (ie moving in the same direction)

iv) passes origin when $x=0$

$\cos 2t + 2 \sin 2t = 0$

$2 \sin 2t = -\cos 2t$

$\tan 2t = -\frac{1}{2}$

$\therefore 2t = n\pi + \tan^{-1}(-\frac{1}{2})$

$t = \frac{n\pi}{2} + \frac{1}{2} \tan^{-1}(-\frac{1}{2})$

$\therefore$ difference in time is $\frac{\pi}{2}$

$[n=0, t_0 = \frac{1}{2} \tan^{-1}(-\frac{1}{2}), n=1, t_1 = \frac{\pi}{2} + \frac{1}{2} \tan^{-1}(-\frac{1}{2}) \therefore t_1 - t_0 = \frac{\pi}{2}]$

NB
 ⊗ Q6b) see Q2 after

Q7 Solutions to Question 7 Ext.1 2009 Trial

(a) $(3+2x)^n = \sum_{k=0}^n {}^nC_k 3^{n-k} (2x)^k$
 $= \sum_{k=0}^n {}^nC_k 3^{n-k} 2^k x^k$

Coefficient of x^5 is ${}^nC_5 3^{n-5} 2^5$

Coefficient of x^6 is ${}^nC_6 3^{n-6} 2^6$

Coefficient of $x^6 = 2 \times$ coefficient of x^5
 ${}^nC_6 3^{n-6} 2^6 = 2 \cdot {}^nC_5 3^{n-5} 2^5$

${}^nC_6 \frac{3^{n-6}}{3^{n-5}} = {}^nC_5$

$\frac{n!}{(n-6)!6!} = \frac{3n!}{(n-5)!5!}$

$\frac{1}{6} = \frac{3}{n-5}$

$n-5=18$

$n=23$

(b) i) $\ddot{x}=0 \quad \ddot{y}=-10$

$\dot{x}=v \cos \alpha \quad \dot{y}=-10t + v \sin \alpha$

$x=vt \cos \alpha \quad y=-5t^2 + vt \sin \alpha$

$\Rightarrow t = \frac{x \sec \alpha}{v}$

$\therefore y = -5 \left(\frac{x \sec \alpha}{v} \right)^2 + x \sin \alpha \sec \alpha$

$y = \frac{-5x^2(1+\tan^2 \alpha)}{v^2} + x \tan \alpha$

(ii) Sub (p,h)

$h = \frac{-5p^2(1+\tan^2 \alpha)}{v^2} + p \tan \alpha$

$h - p \tan \alpha = \frac{-5p^2(1+\tan^2 \alpha)}{v^2}$

$p \tan \alpha - h = \frac{5p^2(1+\tan^2 \alpha)}{v^2}$

$\therefore v^2 = \frac{5p^2(1+\tan^2 \alpha)}{p \tan \alpha - h}$

iii) $v^2 = \frac{5p^2(1+\tan^2 \alpha)}{p \tan \alpha - h}$... ①

Similarly $v^2 = \frac{5q^2(1+\tan^2 \alpha)}{q \tan \alpha - h}$... ②

$\therefore \frac{5p^2(1+\tan^2 \alpha)}{p \tan \alpha - h} = \frac{5q^2(1+\tan^2 \alpha)}{q \tan \alpha - h}$

$\frac{p^2}{p \tan \alpha - h} = \frac{q^2}{q \tan \alpha - h}$

$p^2 q \tan \alpha - p^2 h = q^2 p \tan \alpha - q^2 h$

$p^2 q \tan \alpha - q^2 p \tan \alpha = p^2 h - q^2 h$

$\tan \alpha (p^2 q - q^2 p) = p^2 h - q^2 h$

$\tan \alpha = \frac{h(p-q)(p+q)}{pq(p-q)}$

$\tan \alpha = \frac{h(p+q)}{pq}$

iv) $w^2 = \frac{5p^2(1+\tan^2 \alpha)}{p \tan \alpha - h}$

$\tan \alpha = \frac{h(p+q)}{pq}$

$\therefore w^2 = 5p^2 \left[\frac{1 + \frac{h^2(p+q)^2}{p^2 q^2}}{\frac{h(p+q)}{pq} - h} \right]$

$= 5p^2 \left[\frac{p^2 q^2 + h^2(p+q)^2}{p^2 q^2} \times \frac{pq}{ph(p+q) - hpq} \right]$

$= 5 \left[\frac{p^2 q^2 + h^2(p+q)^2}{q} \times \frac{p}{ph(p+q) - hpq} \right]$

$= 5 \left[\frac{p^2 q^2 + h^2(p+q)^2}{pqh} \right]$

$= \frac{5p^2 q^2}{pqh} + \frac{5h^2(p+q)^2}{pqh}$

$= \frac{5pq}{h} + \frac{5h(p+q)^2}{pqh}$